

Ex. No. : 01

Date:

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A PYTHON PROGRAM TO IMPLEMENT UNIVARIATE, BIVARIATE AND MULTIVARIATE REGRESION

Aim:

Develop a Python program to implement **univariate, bivariate, and multivariate regression** models and analyze the relationship between dependent and independent variables.

Requirements:

Jupyter Notebook with Python, Diabetes.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from sklearn.datasets import load_diabetes

# Load dataset
data = load_diabetes(as_frame=True)
df = data.frame

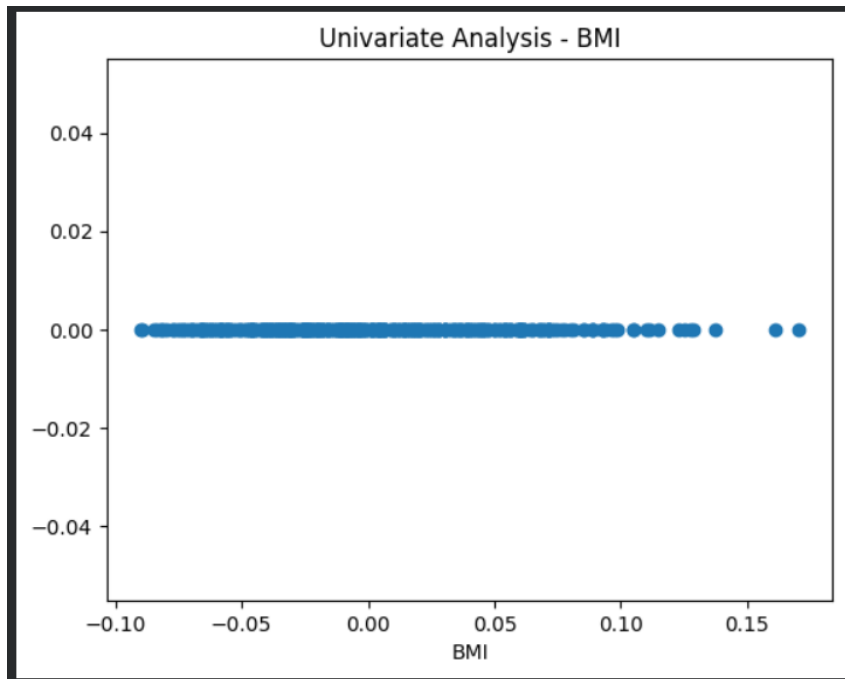
# Rename target column
df.rename(columns={"target": "disease_progression"}, inplace=True)

# Show dataset
df.head()
df.shape

(442, 11)

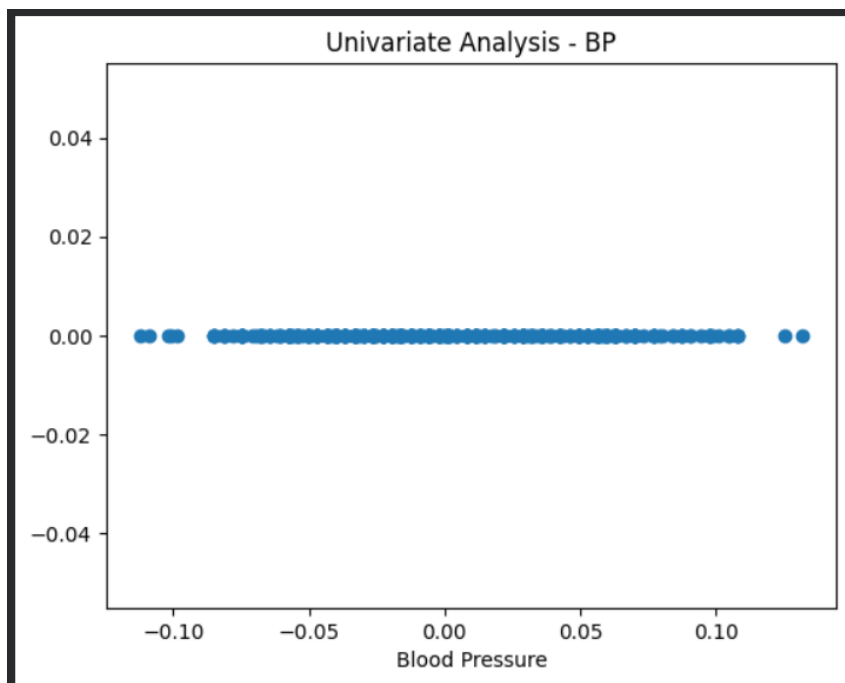
#Univariate for BMI

plt.scatter(df['bmi'], np.zeros_like(df['bmi']))
plt.xlabel('BMI')
plt.title("Univariate Analysis - BMI")
plt.show()
```



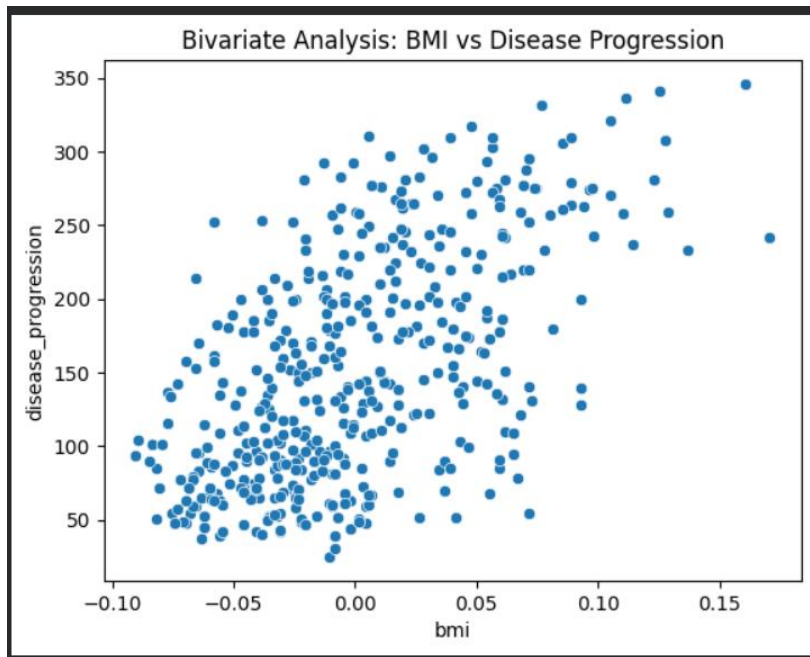
```
#univariate for BP
```

```
plt.scatter(df['bp'], np.zeros_like(df['bp']))  
plt.xlabel('Blood Pressure')  
plt.title("Univariate Analysis - BP")  
plt.show()
```



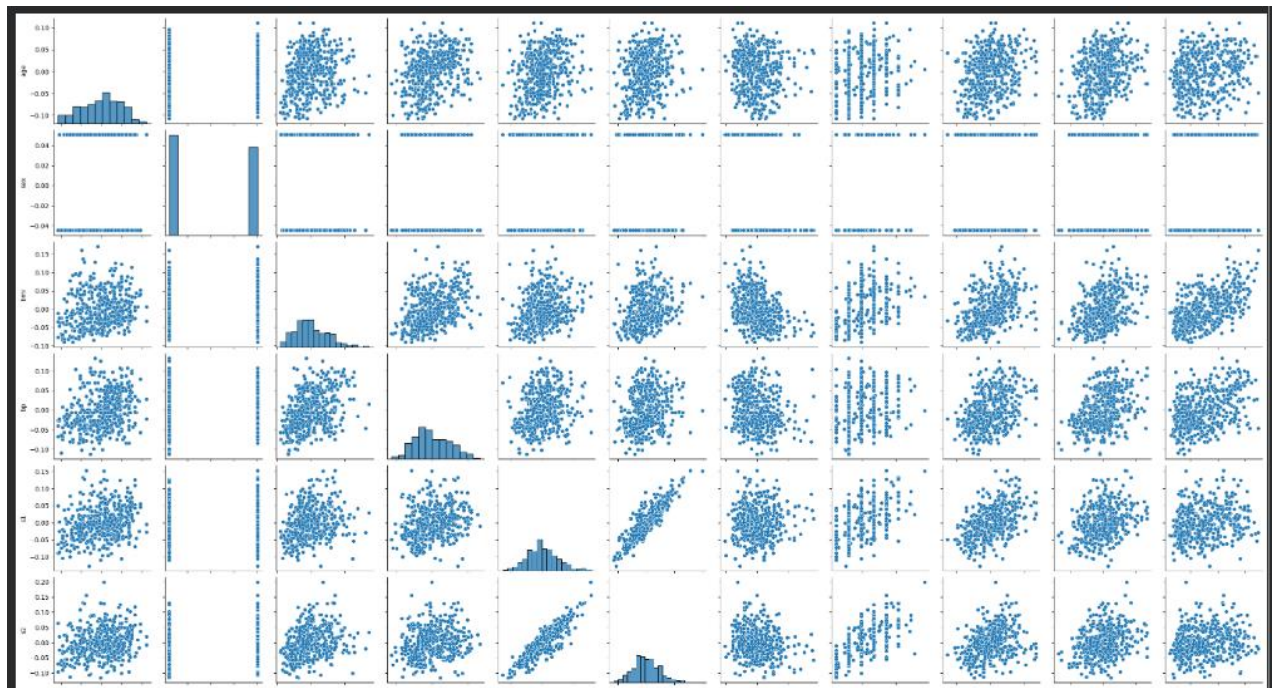
```
# Bivariate Analysis: BMI vs Disease Progression
```

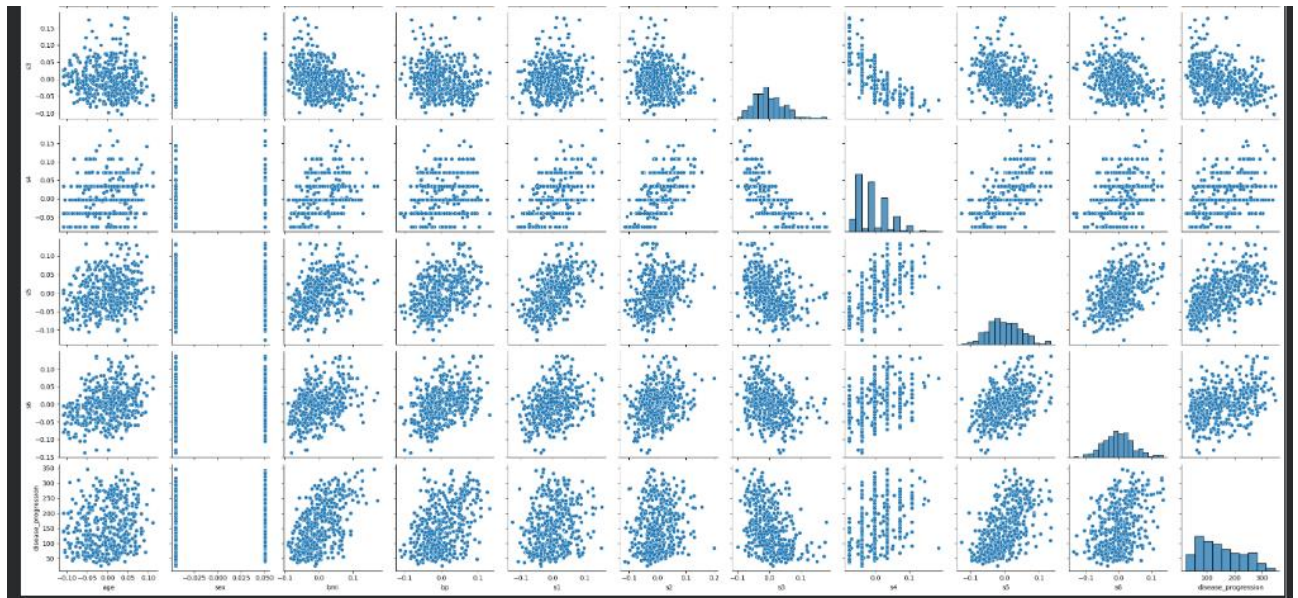
```
sns.scatterplot(x='bmi', y='disease_progression', data=df)  
plt.title("Bivariate Analysis: BMI vs Disease Progression")  
plt.show()
```



multivariate all the features

```
sns.pairplot(df)
plt.show()
```





#Result:

the Python program for univariate, bivariate, and multivariate regression was successfully implemented using the Diabetes.csv dataset in Jupyter Notebook, and the regression models were able to analyze and predict relationships between variables accurately.